Unit 2

Fundamental Concepts of the Digital Economy

Framework Library · 20 reference sheets

LEGEND

DEFINITION	Conceptual definitions and key terms (8 cards)
FRAMEWORK	Strategic frameworks and analytical tools (12 cards)

Dr. Hildegard Haas · EU Business School

E-Business vs. E-Commerce

E-business is the whole organisation; e-commerce is transactions only.

PROFESSIONAL DEFINITION

E-business denotes the application of digital technologies to all business activities — internal operations, supply-chain coordination, customer relationship management, and partner integration. E-commerce is the narrower subset concerned with the digital execution of buying-and-selling transactions over electronic networks.

WHO DEVELOPED IT

Lou Gerstner / IBM (1997) coined the term 'e-business'.

WHY IT WAS DEVELOPED

Kalakota and Whinston (1996) provided the first systematic academic framing in 'Frontiers of Electronic Commerce'.

FIGURE 1 · E-BUSINESS VS. E-COMMERCE

Figure 1. E-Business vs. E-Commerce: E-business is the whole organisation; e-commerce is transactions only.

HOW IT IS USED

when scoping a digital transformation programme to clarify whether the brief is enterprise-wide or transaction-only.

WHEN IT IS USED

in module 1 of any digital strategy course — disambiguating loose vocabulary.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in E-Business Architecture.

REAL-WORLD EXAMPLES

- Amazon: e-commerce front-end (.com checkout) sits inside a vast e-business back-end (AWS, fulfilment, advertising, devices).
- Siemens MindSphere: pure e-business (industrial IoT platform) — almost no consumer e-commerce.
- Etsy: predominantly e-commerce, with light e-business tooling for sellers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, E-Business vs. E-Commerce contributes a clearer, more transferable framing of the dynamic it names — E-business is the whole organisation; e-commerce is transactions only. Its analytical contribution is the explicit recognition that e-business = all business processes mediated digitally (procurement, HR, logistics, marketing).

RELATED FRAMEWORKS IN THIS COURSE

B2B · B2C · C2C · B2G · 2.1.3 | E-Business Infrastructure Stack · 2.1.2 | Digital Business Models (Unit 4)

SOURCES (HARVARD REFERENCING STYLE)

Chaffey, D. and Ellis-Chadwick, F. (2019) Digital Business and E-Commerce Management. 7th edn. Harlow: Pearson.

Kalakota, R. and Whinston, A.B. (1996) Frontiers of Electronic Commerce. Reading, MA: Addison-Wesley.

Laudon, K.C. and Traver, C.G. (2022) E-Commerce: Business, Technology, Society. 17th edn. Harlow: Pearson.

E-Business Infrastructure Stack

SLIBRARY 05 · SHEET 2 / 5

Five layers: hardware → networks → operating systems → applications → business logic.

PROFESSIONAL DEFINITION

A layered reference model that decomposes the technical foundations of any digital business into five interdependent strata. Each layer abstracts complexity from the one above and exposes a clean interface to it.

WHO DEVELOPED IT

Chaffey and Ellis-Chadwick (Digital Business and E-Commerce Management, 1st edn 2002 — refined through the 7th edn 2019).

WHY IT WAS DEVELOPED

The model adapts the OSI seven-layer reference model to commercial e-business.

FIGURE 1 · E-BUSINESS INFRASTRUCTURE STACK

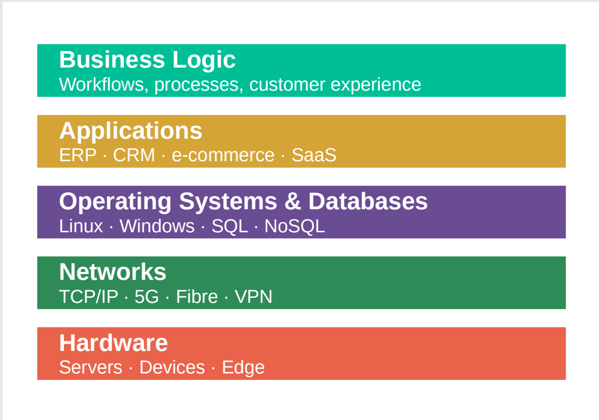

Figure 1. E-Business Infrastructure Stack: Five layers: hardware → networks → operating systems → applications → business logic.

HOW IT IS USED

in capability audits to locate whether a bottleneck is infrastructural, applicational or organisational.

WHEN IT IS USED

when communicating between IT and business stakeholders — assigns ownership by layer.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in E-Business Architecture.

REAL-WORLD EXAMPLES

- Netflix: AWS hardware (L1), CDN networks (L2), Linux+Cassandra (L3), bespoke streaming app (L4), recommendation business logic (L5).
- A typical European retailer: physical data centre (L1), MPLS network (L2), SAP HANA (L3), Hybris commerce (L4), pricing rules (L5).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, E-Business Infrastructure Stack contributes a clearer, more transferable framing of the dynamic it names — Five layers: hardware → networks → operating systems → applications → business logic. Its analytical contribution is the explicit recognition that layer 1 — Hardware: servers, devices, sensors, edge equipment.

RELATED FRAMEWORKS IN THIS COURSE

Cloud: IaaS · PaaS · SaaS · 2.1.5 | API Economy · 2.1.4 | Digital Maturity Model (Unit 4)

SOURCES (HARVARD REFERENCING STYLE)

Chaffey, D. and Ellis-Chadwick, F. (2019) Digital Business and E-Commerce Management. 7th edn. Harlow: Pearson, ch. 3.

Tanenbaum, A.S. and Wetherall, D.J. (2021) Computer Networks. 6th edn. Harlow: Pearson.

Skilton, M. (2015) Building the Digital Enterprise. London: Palgrave Macmillan.

B2B · B2C · C2C · B2G

SLIBRARY 05 · SHEET 3 / 5

Who sells to whom: the four classic e-commerce modes.

PROFESSIONAL DEFINITION

A taxonomy of e-commerce relationships defined by the institutional identity of the seller and the buyer. The four canonical modes — business-to-business, business-to-consumer, consumer-to-consumer, business-to-government — capture the structural variety of digital exchange.

WHO DEVELOPED IT

Codified by Laudon and Traver (1st edn 2001) and now standard in OECD and UNCTAD digital-economy statistics.

WHY IT WAS DEVELOPED

Earlier informal use traceable to U.S. Commerce Department reports of the late 1990s.

FIGURE 1 · B2B · B2C · C2C · B2G

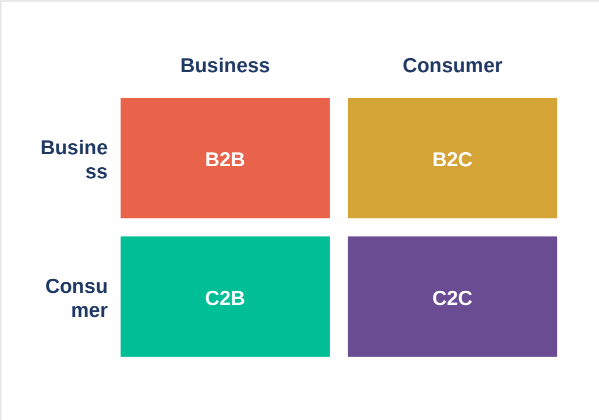

Figure 1. B2B · B2C · C2C · B2G: Who sells to whom: the four classic e-commerce modes.

HOW IT IS USED

when classifying an e-commerce business model in the first analytical pass.

WHEN IT IS USED

in market sizing — each mode has different growth rates and margins.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in E-Business Architecture.

REAL-WORLD EXAMPLES

- Alibaba.com: B2B wholesale.
- Amazon.com: predominantly B2C with a significant B2B (Business) arm.
- eBay, Vinted: C2C marketplaces.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, B2B · B2C · C2C · B2G contributes a clearer, more transferable framing of the dynamic it names — Who sells to whom: the four classic e-commerce modes. Its analytical contribution is the explicit recognition that b2B — sales between enterprises (Alibaba, Salesforce, SAP).

RELATED FRAMEWORKS IN THIS COURSE

E-Business vs. E-Commerce · 2.1.1 | Two-Sided Markets · 2.2.3 | Value Chain → Value Network · 2.3.1

SOURCES (HARVARD REFERENCING STYLE)

Laudon, K.C. and Traver, C.G. (2022) E-Commerce: Business, Technology, Society. 17th edn. Harlow: Pearson.

OECD (2023) OECD Digital Economy Outlook 2023. Paris: OECD Publishing.

UNCTAD (2024) Digital Economy Report 2024. Geneva: United Nations.

API Economy

SLIBRARY 05 · SHEET 4 / 5

Software eating the world — via Application Programming Interfaces.

PROFESSIONAL DEFINITION

An economic configuration in which firms expose, consume and monetise discrete software capabilities through standardised programmatic interfaces. APIs become products in their own right, enabling composable architectures, partner ecosystems and new revenue streams.

WHO DEVELOPED IT

Conceptual foundations: Jeff Bezos's 2002 'API mandate' at Amazon.

WHY IT WAS DEVELOPED

Popularised by Iyer and Subramaniam (Harvard Business Review, 2015) and the work of Daniel Jacobson (Netflix). Marc Andreessen's 'Why Software Is Eating the World' (Wall Street Journal, 2011) provides the broader economic frame.

FIGURE 1 · API ECONOMY

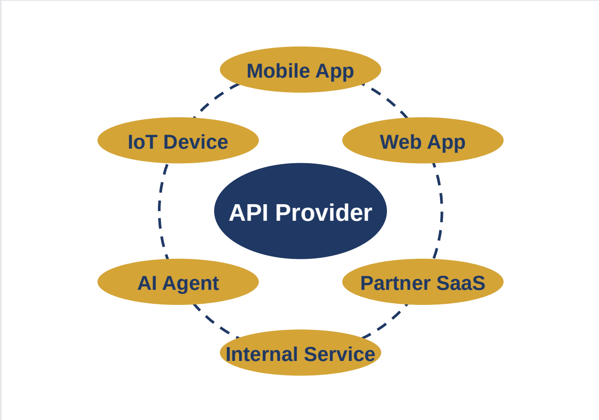

Figure 1. API Economy: Software eating the world — via Application Programming Interfaces.

HOW IT IS USED

when assessing whether a firm should productise a capability or keep it internal.

WHEN IT IS USED

in platform-strategy work — APIs are the technical instantiation of openness decisions.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in E-Business Architecture.

REAL-WORLD EXAMPLES

- Stripe: payments-as-an-API. ~$1.4 trillion in payment volume processed (2024).
- Twilio: communications APIs (SMS, voice, video).
- OpenAI: AI inference monetised through API calls — the canonical AI-era API business.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, API Economy contributes a clearer, more transferable framing of the dynamic it names — Software eating the world — via Application Programming Interfaces. Its analytical contribution is the explicit recognition that provider exposes a capability (payment, mapping, identity, AI inference) via a documented interface.

RELATED FRAMEWORKS IN THIS COURSE

- E-Business Infrastructure Stack · 2.1.2
- Cloud: IaaS · PaaS · SaaS · 2.1.5
- Platform Strategy (Unit 3)

SOURCES (HARVARD REFERENCING STYLE)

Iyer, B. and Subramaniam, M. (2015) 'The strategic value of APIs', Harvard Business Review, 7 January.

Andreessen, M. (2011) 'Why software is eating the world', Wall Street Journal, 20 August.

Jacobson, D., Brail, G. and Woods, D. (2012) APIs: A Strategy Guide. Sebastopol, CA: O'Reilly.

Cloud: IaaS · PaaS · SaaS

Three levels of what the cloud provider manages for you.

PROFESSIONAL DEFINITION

A canonical service-model taxonomy distinguishing the depth at which a cloud provider takes operational responsibility — from raw infrastructure (IaaS), through managed runtime (PaaS), to fully delivered software (SaaS).

WHO DEVELOPED IT

Formalised by the U.S.

WHY IT WAS DEVELOPED

National Institute of Standards and Technology — Mell and Grance, 'The NIST Definition of Cloud Computing' (Special Publication 800-145, September 2011). Now the global de-facto standard.

FIGURE 1 · CLOUD: IAAS · PAAS · SAAS

Figure 1. Cloud: IaaS · PaaS · SaaS: Three levels of what the cloud provider manages for you.

HOW IT IS USED

in technology-sourcing decisions — explicitly trades control against operational burden.

WHEN IT IS USED

in cost modelling — TCO calculations differ sharply across the three.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in E-Business Architecture.

REAL-WORLD EXAMPLES

- IaaS: AWS EC2, Microsoft Azure VMs, Google Compute Engine.
- PaaS: Heroku, Vercel, Google App Engine, AWS Elastic Beanstalk.
- SaaS: Salesforce, Microsoft 365, Workday, Slack, Notion.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Cloud: IaaS · PaaS · SaaS contributes a clearer, more transferable framing of the dynamic it names — Three levels of what the cloud provider manages for you. Its analytical contribution is the explicit recognition that iaaS — provider supplies virtualised compute, storage, networking. Customer manages OS upward.

RELATED FRAMEWORKS IN THIS COURSE

E-Business Infrastructure Stack · 2.1.2 | API Economy · 2.1.4 | Subscription Economy · 2.3.3

SOURCES (HARVARD REFERENCING STYLE)

Mell, P. and Grance, T. (2011) The NIST Definition of Cloud Computing. NIST Special Publication 800-145. Gaithersburg, MD: NIST.

Marston, S. et al. (2011) 'Cloud computing — the business perspective', Decision Support Systems, 51(1), pp. 176–189.

Armbrust, M. et al. (2010) 'A view of cloud computing', Communications of the ACM, 53(4), pp. 50–58.

Metcalfe's Law

Network value grows as n^2 (number of users squared).

PROFESSIONAL DEFINITION

An empirical heuristic stating that the value of a communications network is proportional to the square of the number of connected users (n^2), because each new user can potentially connect with every existing user.

WHO DEVELOPED IT

Robert Metcalfe — co-inventor of Ethernet at Xerox PARC, founder of 3Com.

WHY IT WAS DEVELOPED

Articulated informally in the early 1980s; first formalised by George Gilder ('Metcalfe's Law and Legacy', Forbes ASAP, 1993).

FIGURE 1 · METCALFE'S LAW

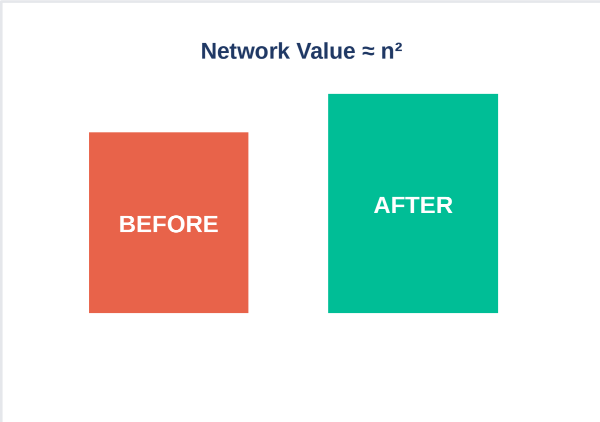

Figure 1. Metcalfe's Law: Network value grows as n^2 (number of users squared).

HOW IT IS USED

as the first lens for any growth-stage platform business — how much does each new user actually add?

WHEN IT IS USED

in venture diligence — distinguishes networks that compound from those that merely scale linearly.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Networks.

REAL-WORLD EXAMPLES

- Facebook: explicit Metcalfe-style growth from 2004–2012; valuation tracked user² across early IPO years.
- WhatsApp: $19bn Facebook acquisition (2014) justified explicitly on 450m-user network value.
- Fax machines: classic textbook case — useless alone, indispensable at scale, obsolete when displaced.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Metcalfe's Law contributes a clearer, more transferable framing of the dynamic it names — Network value grows as n^2 (number of users squared). Its analytical contribution is the explicit recognition that n users → $n(n-1)/2$ possible pairwise connections → ≈ n^2 value.

RELATED FRAMEWORKS IN THIS COURSE

Direct vs. Indirect Network Effects · 2.2.2 | Two-Sided Markets · 2.2.3 | Tipping Point · Winner-Takes-All · 2.2.5

SOURCES (HARVARD REFERENCING STYLE)

Gilder, G. (1993) 'Metcalfe's Law and legacy', Forbes ASAP, 13 September.

Briscoe, B., Odlyzko, A. and Tilly, B. (2006) 'Metcalfe's Law is wrong', IEEE Spectrum, 43(7), pp. 34–39.

Shapiro, C. and Varian, H.R. (1999) Information Rules. Boston, MA: Harvard Business School Press.

Direct vs. Indirect Network Effects

SLIBRARY 06 · SHEET 2 / 5

Same-side benefit vs. cross-side benefit.

PROFESSIONAL DEFINITION

A distinction between two mechanisms by which networks create value. Direct (same-side) effects arise when more users on one side make the platform more valuable to that same side. Indirect (cross-side) effects arise when more users on one side make the platform more valuable to a different side.

WHO DEVELOPED IT

Conceptual foundations in Katz and Shapiro ('Network externalities, competition, and compatibility', American Economic Review, 1985).

WHY IT WAS DEVELOPED

Refined for two-sided platforms by Rochet and Tirole (2003) and popularised by Eisenmann, Parker and Van Alstyne (Harvard Business Review, 2006).

FIGURE 1 · DIRECT VS. INDIRECT NETWORK EFFECTS

Figure 1. Direct vs. Indirect Network Effects: Same-side benefit vs. cross-side benefit.

HOW IT IS USED

when designing a platform launch — direct effects suggest single-sided growth tactics; indirect effects demand chicken-and-egg solutions.

WHEN IT IS USED

in competitive analysis — identifies which side of a market is the constraint.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Networks.

REAL-WORLD EXAMPLES

- Direct: WhatsApp, LinkedIn (within a profession), multiplayer games.
- Indirect: Uber (riders↔drivers), Visa (cardholders↔merchants), Airbnb (guests↔hosts), iOS (users↔developers).
- Mixed: YouTube (creator-viewer indirect + viewer-viewer direct via comments and recommendations).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Direct vs. Indirect Network Effects contributes a clearer, more transferable framing of the dynamic it names — Same-side benefit vs. cross-side benefit. Its analytical contribution is the explicit recognition that direct: users on side A benefit from more users on side A. Telephones, WhatsApp, Slack within a firm.

RELATED FRAMEWORKS IN THIS COURSE

Metcalfe's Law · 2.2.1 | Two-Sided Markets · 2.2.3 | Tipping Point · Winner-Takes-All · 2.2.5

SOURCES (HARVARD REFERENCING STYLE)

Katz, M.L. and Shapiro, C. (1985) 'Network externalities, competition, and compatibility', American Economic Review, 75(3), pp. 424–440.

Eisenmann, T., Parker, G. and Van Alstyne, M.W. (2006) 'Strategies for two-sided markets', Harvard Business Review, October, pp. 92–101.

Parker, G., Van Alstyne, M. and Choudary, S.P. (2016) Platform Revolution. New York: W.W. Norton.

Two-Sided Markets

Rochet & Tirole: platforms solve a coordination problem between two distinct user groups.

PROFESSIONAL DEFINITION

An economic model of markets in which a platform intermediary brings together two (or more) distinct user groups whose value to each other is realised only through the platform. The platform sets a price structure — not just a price level — to balance participation on both sides.

WHO DEVELOPED IT

Jean-Charles Rochet and Jean Tirole, 'Platform competition in two-sided markets' (Journal of the European Economic Association, 2003).

WHY IT WAS DEVELOPED

Tirole was awarded the 2014 Nobel Prize in Economics partly for this work.

FIGURE 1 · TWO-SIDED MARKETS

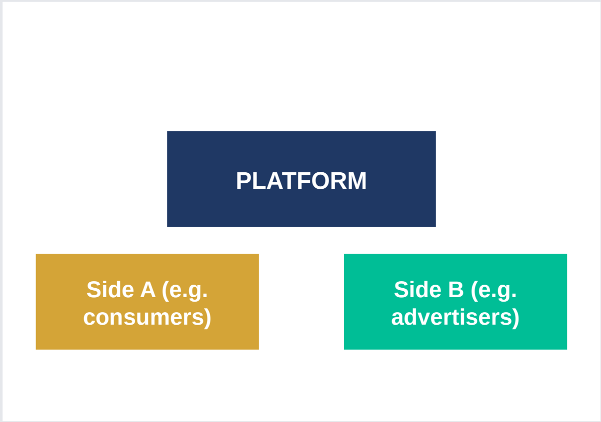

Figure 1. Two-Sided Markets: Rochet & Tirole: platforms solve a coordination problem between two distinct user groups.

HOW IT IS USED

when designing or auditing a platform's pricing — surfaces non-obvious price-structure choices.

WHEN IT IS USED

in regulatory analysis — competition authorities now treat two-sided pricing as standard.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Networks.

REAL-WORLD EXAMPLES

- Google Search: free for users (subsidy side), paid by advertisers (money side).
- Adobe (historic): students discounted, professionals at full price — a same-product two-tier strategy.
- Visa & Mastercard: textbook Rochet–Tirole case (cardholders subsidised, merchants pay).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Two-Sided Markets contributes a clearer, more transferable framing of the dynamic it names — Rochet & Tirole: platforms solve a coordination problem between two distinct user groups. Its analytical contribution is the explicit recognition that the platform's challenge is not setting one price but a price structure across both sides.

RELATED FRAMEWORKS IN THIS COURSE

Direct vs. Indirect Network Effects · 2.2.2 | Platform Strategy (Unit 3) | Price Discrimination · 2.4.4

SOURCES (HARVARD REFERENCING STYLE)

Rochet, J.-C. and Tirole, J. (2003) 'Platform competition in two-sided markets', Journal of the European Economic Association, 1(4), pp. 990–1029.

Rochet, J.-C. and Tirole, J. (2006) 'Two-sided markets: a progress report', RAND Journal of Economics, 37(3), pp. 645–667.

Evans, D.S. and Schmalensee, R. (2016) Matchmakers: The New Economics of Multisided Platforms. Boston, MA: Harvard Business Review Press.

Connection Density (MIT IDE 2025)

MIT IDE update: not more users, but denser connections drive value.

PROFESSIONAL DEFINITION

A refinement of classical network-effect theory holding that, beyond a threshold, the marginal value of an additional user falls — while the marginal value of an additional meaningful connection (interaction-weighted edge) keeps rising. Value tracks the density and quality of edges, not the count of nodes.

WHO DEVELOPED IT

MIT Initiative on the Digital Economy (IDE) — synthesised in MIT IDE Annual Report 2025 (Brynjolfsson, McAfee and colleagues), drawing on prior work by Tucker (2018) and on AI-mediated network analyses by Aral and Walker.

WHY IT WAS DEVELOPED

The framework was developed to give practitioners a precise, transferable vocabulary for the dynamic captured in its tagline — replacing the looser pre-existing terms used in earlier industry and academic discourse.

FIGURE 1 · CONNECTION DENSITY (MIT IDE 2025)

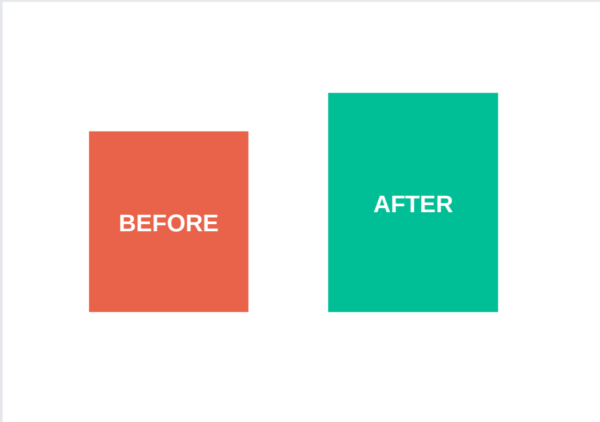

Figure 1. Connection Density (MIT IDE 2025): MIT IDE update: not more users, but denser connections drive value.

HOW IT IS USED

in growth-stage analytics where raw user numbers plateau — shifts attention to interaction depth.

WHEN IT IS USED

when justifying personalisation and recommender investment — they raise edge density.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Networks.

REAL-WORLD EXAMPLES

- TikTok: per-session engagement multiples higher than predecessor platforms despite fewer users.
- LinkedIn (post-AI features 2023–2025): connection-density growth without proportionate user growth.
- Slack vs. Email: same number of contacts, vastly higher interaction density.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Connection Density (MIT IDE 2025) contributes a clearer, more transferable framing of the dynamic it names — MIT IDE update: not more users, but denser connections drive value. Its analytical contribution is the explicit recognition that classical Metcalfe assumes all edges equally valuable — empirically false at scale.

RELATED FRAMEWORKS IN THIS COURSE

Metcalfe's Law · 2.2.1 | Direct vs. Indirect Network Effects · 2.2.2 | AI & Personalisation (Unit 4)

SOURCES (HARVARD REFERENCING STYLE)

Brynjolfsson, E. and McAfee, A. (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Initiative on the Digital Economy.

Tucker, C. (2018) 'Network effects and market power: what have we learned in the last decade?', Antitrust, 32(2), pp. 72–79.

Aral, S. (2020) The Hype Machine. New York: Currency.

Tipping Point · Winner-Takes-All

At critical mass, the market tips to one platform.

PROFESSIONAL DEFINITION

A market dynamic in which strong network effects, low marginal costs and limited differentiation cause user adoption to converge on a single dominant platform once a critical-mass threshold is crossed. Smaller competitors enter a death spiral as users rationally migrate to the larger network.

WHO DEVELOPED IT

Conceptual ancestry: Schelling, 'Micromotives and Macrobehavior' (1978), and Arthur, 'Increasing returns and path dependence' (1989).

WHY IT WAS DEVELOPED

Popularised in business by Shapiro and Varian (Information Rules, 1999) and applied to platforms by Eisenmann et al. (2006).

FIGURE 1 · TIPPING POINT · WINNER-TAKES-ALL

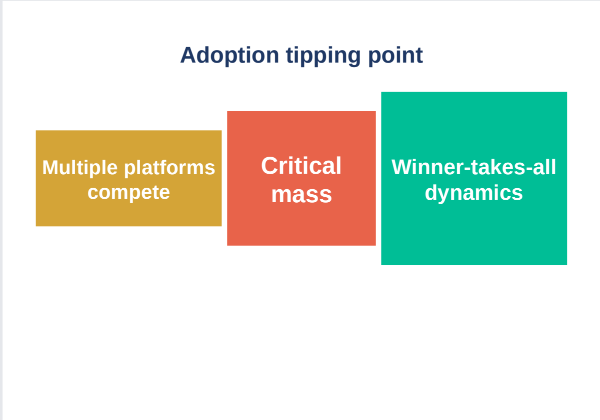

Figure 1. Tipping Point · Winner-Takes-All: At critical mass, the market tips to one platform.

HOW IT IS USED

when assessing market structure — predicts whether duopoly is sustainable.

WHEN IT IS USED

in regulatory and antitrust analysis — diagnoses concentration risk.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Networks.

REAL-WORLD EXAMPLES

- Search: Google ≈90% global share — classic winner-takes-most.
- Operating systems: Windows on PC, iOS+Android on mobile — duopolistic tip.
- Ride-hailing: Uber/Lyft U.S.; Didi China — geographic tipping with multi-homing.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Tipping Point · Winner-Takes-All contributes a clearer, more transferable framing of the dynamic it names — At critical mass, the market tips to one platform. Its analytical contribution is the explicit recognition that below the tipping point, multiple platforms can coexist.

RELATED FRAMEWORKS IN THIS COURSE

Metcalfe's Law · 2.2.1 | Lock-In & Switching Costs · 2.4.5 | Platform Liability & Regulation (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

Arthur, W.B. (1989) 'Competing technologies, increasing returns, and lock-in by historical events', The Economic Journal, 99(394), pp. 116–131.

Shapiro, C. and Varian, H.R. (1999) Information Rules. Boston, MA: Harvard Business School Press.

Parker, G., Van Alstyne, M. and Choudary, S.P. (2016) Platform Revolution. New York: W.W. Norton.

Value Chain → Value Network

SLIBRARY 07 · SHEET 1 / 5

Porter's linear chain replaced by hub-and-spoke value networks.

PROFESSIONAL DEFINITION

A reconceptualisation of value creation in digital settings, in which Porter's linear sequence of primary activities (inbound logistics → operations → outbound logistics → marketing → service) gives way to a non-linear network of interacting actors orchestrated by a digital platform.

WHO DEVELOPED IT

Porter's Value Chain — Competitive Advantage (1985).

WHY IT WAS DEVELOPED

Reformulation for the network economy: Stabell and Fjeldstad ('Configuring value for competitive advantage', Strategic Management Journal, 1998) and Tapscott (Digital Capital, 2000). Revisited by Iansiti and Lakhani (Competing in the Age of AI, 2020).

FIGURE 1 · VALUE CHAIN → VALUE NETWORK

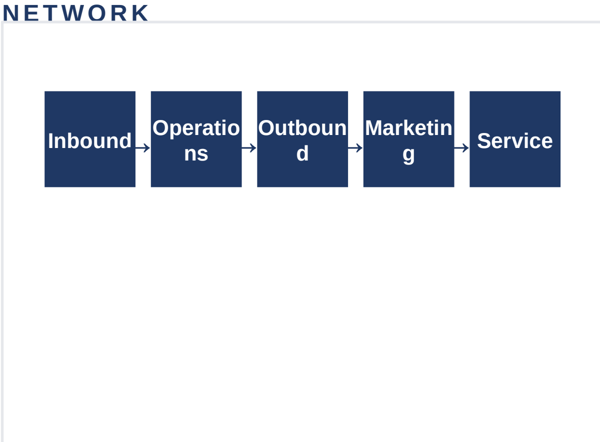

Figure 1. Value Chain → Value Network: Porter's linear chain replaced by hub-and-spoke value networks.

HOW IT IS USED

when analysing why platform-orchestrators outperform vertically-integrated incumbents.

WHEN IT IS USED

in business-model redesign — surfaces opportunities to disintermediate or to become an orchestrator.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Value Creation.

REAL-WORLD EXAMPLES

- Apple App Store: Apple does not write the apps — it orchestrates a network of 30m+ developers.
- Airbnb: zero hotel real estate, 7m+ listings — pure network orchestration.
- Tesla vs. legacy auto: legacy = chain (suppliers → assembly → dealers); Tesla = network (over-the-air updates, data feedback, integrated charging).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Value Chain → Value Network contributes a clearer, more transferable framing of the dynamic it names — Porter's linear chain replaced by hub-and-spoke value networks. Its analytical contribution is the explicit recognition that linear value chain: input → activity → output, value added stage by stage.

RELATED FRAMEWORKS IN THIS COURSE

Two-Sided Markets · 2.2.3 | Platform Strategy (Unit 3) | Data as Strategic Asset · 2.3.4

SOURCES (HARVARD REFERENCING STYLE)

Porter, M.E. (1985) Competitive Advantage. New York: Free Press.

Stabell, C.B. and Fjeldstad, Ø.D. (1998) 'Configuring value for competitive advantage', Strategic Management Journal, 19(5), pp. 413–437.

Iansiti, M. and Lakhani, K.R. (2020) Competing in the Age of AI. Boston, MA: Harvard Business Review Press.

Freemium Model

Free for most users · premium features for the paying few.

PROFESSIONAL DEFINITION

A revenue-model archetype in which a basic version of a digital product is provided at no charge to acquire a large user base, while a premium tier — accessed by a small minority — generates the revenue. Variable costs near zero make this economically viable.

WHO DEVELOPED IT

Term coined by Jarid Lukin and popularised by venture capitalist Fred Wilson on his blog 'A VC' (2006).

WHY IT WAS DEVELOPED

Theoretical foundations in Anderson, 'Free: The Future of a Radical Price' (2009).

FIGURE 1 · FREEMIUM MODEL

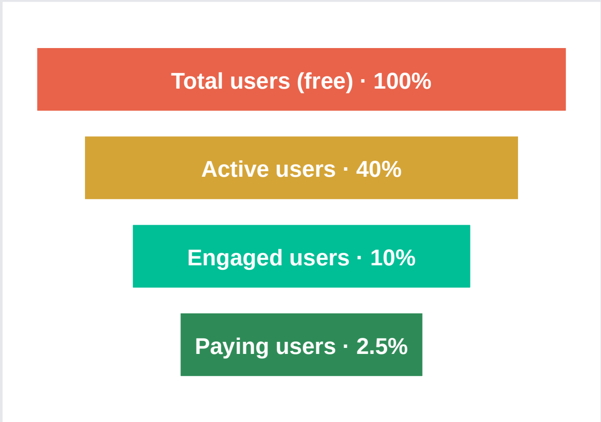

Figure 1. Freemium Model: Free for most users · premium features for the paying few.

HOW IT IS USED

when launching a new digital product seeking rapid user acquisition.

WHEN IT IS USED

when network effects mean free users themselves create value beyond their direct revenue.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Value Creation.

REAL-WORLD EXAMPLES

- Spotify: ~575m monthly active users, ~239m premium subscribers (Q2 2024) — ~42% conversion, exceptional.
- Dropbox: free 2GB tier; premium tiers from $9.99/mo. Conversion rate ~4%.
- LinkedIn: free networking; premium for recruiters and job-seekers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Freemium Model contributes a clearer, more transferable framing of the dynamic it names — Free for most users · premium features for the paying few. Its analytical contribution is the explicit recognition that free tier acquires users at zero per-user cost — only marginal infrastructure.

RELATED FRAMEWORKS IN THIS COURSE

Subscription Economy · 2.3.3 | Versioning Strategy · 2.4.3 | Long Tail · 2.3.5

SOURCES (HARVARD REFERENCING STYLE)

Anderson, C. (2009) Free: The Future of a Radical Price. New York: Hyperion.

Kumar, V. (2014) 'Making freemium work', Harvard Business Review, May.

Tzuo, T. and Weisert, G. (2018) Subscribed. New York: Portfolio.

Subscription Economy

Recurring revenue replaces one-time transactions.

PROFESSIONAL DEFINITION

An economic configuration in which firms generate revenue through ongoing, time-based access to a service rather than one-time sales of a product. The customer relationship — not the transaction — becomes the locus of value.

WHO DEVELOPED IT

Conceptual roots in services marketing (Berry, 1983).

WHY IT WAS DEVELOPED

Popularised for the digital era by Tien Tzuo (founder of Zuora) — Subscribed: Why the Subscription Model Will Be Your Company's Future and What to Do About It (Portfolio, 2018).

FIGURE 1 · SUBSCRIPTION ECONOMY

Subscription unit economics

ARR (Annual Recurring Revenue)	*The headline north-star number*
MRR	*Monthly equivalent*
Gross churn	*% of customers lost per period*
Net Revenue Retention	*Expansion + retention − churn*
CAC payback	*Months to recover acquisition cost*

Figure 1. Subscription Economy: Recurring revenue replaces one-time transactions.

HOW IT IS USED

when designing or transitioning a business model toward recurring revenue.

WHEN IT IS USED

in valuation work — subscription firms trade on revenue multiples driven by growth, churn, and NRR.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Value Creation.

REAL-WORLD EXAMPLES

- Netflix: ~283m subscribers (Q3 2024) — paradigmatic subscription business.
- Adobe: switched from boxed Creative Suite to Creative Cloud (2013) — share price 10× over the next decade.
- Microsoft 365, Salesforce, ServiceNow: enterprise SaaS — the subscription model at industrial scale.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Subscription Economy contributes a clearer, more transferable framing of the dynamic it names — Recurring revenue replaces one-time transactions. Its analytical contribution is the explicit recognition that recurring revenue replaces episodic transactions — improves predictability and lifetime value.

RELATED FRAMEWORKS IN THIS COURSE

Freemium Model · 2.3.2 | Cloud: SaaS · 2.1.5 | Lock-In & Switching Costs · 2.4.5

SOURCES (HARVARD REFERENCING STYLE)

Tzuo, T. and Weisert, G. (2018) Subscribed. New York: Portfolio.

Baxter, R.K. (2020) The Forever Transaction. New York: McGraw-Hill.

McKinsey & Company (2021) 'Thinking inside the subscription box'.

Data as Strategic Asset

SLIBRARY 07 · SHEET 4 / 5

'The new oil' — with important differences from oil.

PROFESSIONAL DEFINITION

The proposition that data, accumulated through digital operations, is itself a strategic resource: it confers competitive advantage, can be monetised directly or indirectly, and exhibits properties (non-rivalry, increasing returns) distinct from physical commodities.

WHO DEVELOPED IT

Metaphor 'data is the new oil' coined by Clive Humby (Tesco Clubcard architect) in 2006 and amplified by The Economist (2017).

WHY IT WAS DEVELOPED

Strategic-asset theorisation in Mayer-Schönberger and Ramge (Reinventing Capitalism in the Age of Big Data, 2018) and Iansiti and Lakhani (2020).

FIGURE 1 · DATA AS STRATEGIC ASSET

Data vs. oil — four differences

Use	Consumed once	Reusable indefinitely
Returns	Diminishing	Increasing with scale
Refining	Standardised	Highly context-specific
Regulation	Environmental	Privacy & rights-based

Figure 1. Data as Strategic Asset: 'The new oil' — with important differences from oil.

HOW IT IS USED

when assessing competitive moats in data-intensive industries.

WHEN IT IS USED

in M&A diligence — data assets are increasingly the core valuation driver.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Digital Value Creation.

REAL-WORLD EXAMPLES

- Google: query data → ranking improvement → more queries — the canonical data flywheel.
- Tesla: 8bn+ km of real-world driving telemetry — central to autonomy moat.
- Bloomberg Terminal: financial data + analytics, $25k+/seat/year, ~325k subscribers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Data as Strategic Asset contributes a clearer, more transferable framing of the dynamic it names — 'The new oil' — with important differences from oil. Its analytical contribution is the explicit recognition that non-rival: same data can serve unlimited uses simultaneously — unlike oil.

RELATED FRAMEWORKS IN THIS COURSE

Connection Density · 2.2.4 | Surveillance Capitalism (Unit 5) | GDPR & Data Rights (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

Mayer-Schönberger, V. and Ramge, T. (2018) Reinventing Capitalism in the Age of Big Data. New York: Basic Books.

Iansiti, M. and Lakhani, K.R. (2020) Competing in the Age of AI. Boston, MA: Harvard Business Review Press.

The Economist (2017) 'The world's most valuable resource is no longer oil, but data', 6 May.

The Long Tail

Anderson: infinite shelf space monetises niche demand.

PROFESSIONAL DEFINITION

An empirical regularity, formalised for digital markets, holding that a large share of total demand is concentrated in a long tail of low-popularity items, which becomes profitable to serve only when the cost of inventory and distribution approaches zero — as is typical online.

WHO DEVELOPED IT

Chris Anderson — 'The Long Tail', Wired magazine (October 2004), expanded into the book The Long Tail: Why the Future of Business is Selling Less of More (Hyperion, 2006).

WHY IT WAS DEVELOPED

Statistical foundation in Pareto distributions and Zipf's law.

FIGURE 1 · THE LONG TAIL

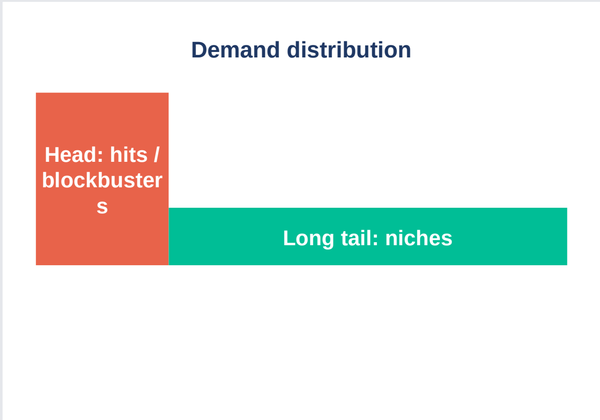

Figure 1. The Long Tail: Anderson: infinite shelf space monetises niche demand.

HOW IT IS USED

when assessing business viability in markets with very large catalogues.

WHEN IT IS USED

when designing recommendation strategy — surfacing the tail is the key UX problem.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Value Creation.

REAL-WORLD EXAMPLES

- Amazon: estimates suggest ~30% of revenue from titles a typical bookstore could not stock.
- Spotify: catalogue of 100m+ tracks; the long tail of obscure artists generates meaningful aggregate revenue.
- Netflix DVD-era: the original Long Tail case — 60% of rentals were films a Blockbuster store could not carry.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, The Long Tail contributes a clearer, more transferable framing of the dynamic it names — Anderson: infinite shelf space monetises niche demand. Its analytical contribution is the explicit recognition that demand follows a power law: a few hits + a long tail of niches.

RELATED FRAMEWORKS IN THIS COURSE

Freemium Model · 2.3.2 | Zero Marginal Cost · 2.4.1 | Versioning Strategy · 2.4.3

SOURCES (HARVARD REFERENCING STYLE)

Anderson, C. (2006) The Long Tail: Why the Future of Business is Selling Less of More. New York: Hyperion.

Brynjolfsson, E., Hu, Y. and Smith, M.D. (2010) 'The longer tail: the changing shape of Amazon's sales distribution curve', Significance, 7(2), pp. 67–71.

Anderson, C. (2004) 'The Long Tail', Wired, October.

Zero Marginal Cost

High to produce the first copy · ~zero to produce the next.

PROFESSIONAL DEFINITION

An economic property of digital goods whereby the cost of producing one additional unit, after the first copy is created, approaches zero. The cost structure is dominated by sunk fixed costs (development, content production) rather than variable production costs.

WHO DEVELOPED IT

Foundational analysis in Shapiro and Varian, Information Rules (1999, ch.1).

WHY IT WAS DEVELOPED

Economic-history extension in Rifkin, The Zero Marginal Cost Society (2014). Extended to AI inference economics in MIT IDE Annual Report 2025.

FIGURE 1 · ZERO MARGINAL COST

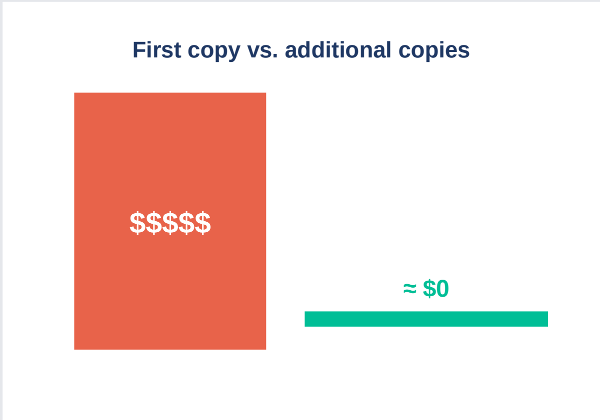

Figure 1. Zero Marginal Cost: High to produce the first copy · ~zero to produce the next.

HOW IT IS USED

in pricing strategy — explains why digital goods can use freemium, versioning, and discrimination.

WHEN IT IS USED

in business-case analysis — surfaces enormous operating-leverage effects at scale.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Information Goods.

REAL-WORLD EXAMPLES

- Software: Microsoft Windows costs billions to develop; each additional licence costs cents to deliver.
- Streaming media: Netflix's content cost is fixed; serving an additional viewer is near-zero bandwidth cost.
- Counter-example: AI inference (e.g. GPT-4) — each query costs measurable compute, breaking the classical model.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Zero Marginal Cost contributes a clearer, more transferable framing of the dynamic it names — High to produce the first copy · ~zero to produce the next. Its analytical contribution is the explicit recognition that first copy: large up-front investment (R&D, content, infrastructure).

RELATED FRAMEWORKS IN THIS COURSE

Freemium Model · 2.3.2 | Long Tail · 2.3.5 | Versioning Strategy · 2.4.3

SOURCES (HARVARD REFERENCING STYLE)

Shapiro, C. and Varian, H.R. (1999) Information Rules. Boston, MA: Harvard Business School Press.

Rifkin, J. (2014) The Zero Marginal Cost Society. New York: Palgrave Macmillan.

Brynjolfsson, E. and McAfee, A. (2025) MIT IDE Annual Report 2025. Cambridge, MA: MIT Initiative on the Digital Economy.

Experience Goods

SLIBRARY 08 · SHEET 2 / 5

Quality unknowable until consumption — and often after.

PROFESSIONAL DEFINITION

A category of goods whose qualities cannot be assessed by inspection prior to purchase: they must be experienced. Most information goods (books, films, software) are experience goods; some are 'credence goods' whose quality is hard to verify even after consumption.

WHO DEVELOPED IT

Phillip Nelson, 'Information and consumer behaviour' (Journal of Political Economy, 1970).

WHY IT WAS DEVELOPED

Extended to credence goods by Darby and Karni (1973). Applied to digital goods systematically by Shapiro and Varian (1999).

FIGURE 1 · EXPERIENCE GOODS

Figure 1. Experience Goods: Quality unknowable until consumption — and often after.

HOW IT IS USED

when designing marketing strategy for any new digital product.

WHEN IT IS USED

when explaining why ratings, reviews, and trial mechanics dominate digital UX.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Information Goods.

REAL-WORLD EXAMPLES

- Spotify: listeners judge a song by playing it — pure experience good.
- Steam refund policy: 2-hour playable trial converts game from experience to quasi-search good.
- Books on Amazon: 'Look Inside' previews reduce experience-good frictions.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Experience Goods contributes a clearer, more transferable framing of the dynamic it names — Quality unknowable until consumption — and often after. Its analytical contribution is the explicit recognition that search goods: quality observable before purchase (e.g. apples).

RELATED FRAMEWORKS IN THIS COURSE

Versioning Strategy · 2.4.3 | Freemium Model · 2.3.2 | Trust & Reputation (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

Nelson, P. (1970) 'Information and consumer behaviour', Journal of Political Economy, 78(2), pp. 311–329.

Darby, M.R. and Karni, E. (1973) 'Free competition and the optimal amount of fraud', Journal of Law and Economics, 16(1), pp. 67–88.

Shapiro, C. and Varian, H.R. (1999) Information Rules. Boston, MA: Harvard Business School Press.

Versioning Strategy

Multiple versions at different prices — let customers self-sort.

PROFESSIONAL DEFINITION

A pricing-and-product strategy in which a firm offers several variants of essentially the same digital good, each with different features and prices, designed so that different customer segments self-select into the version that matches their willingness to pay.

WHO DEVELOPED IT

Formalised by Shapiro and Varian, Information Rules (1999, ch.3) drawing on Pigou's price-discrimination theory (1920).

WHY IT WAS DEVELOPED

Widely operationalised in software industry from the late 1990s.

FIGURE 1 · VERSIONING STRATEGY

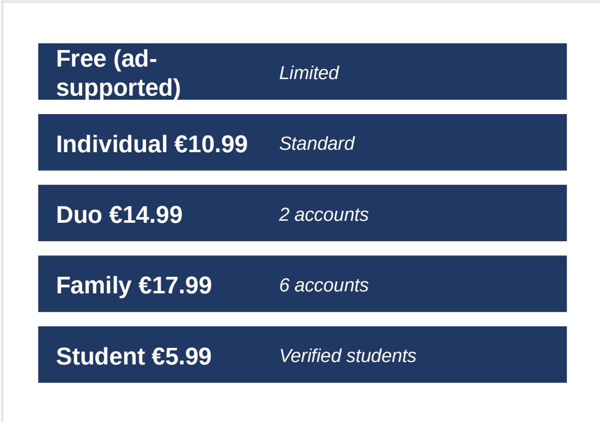

Figure 1. Versioning Strategy: Multiple versions at different prices — let customers self-sort.

HOW IT IS USED

when designing product packaging and pricing for a digital good with heterogeneous demand.

WHEN IT IS USED

in pricing reviews — surfaces opportunities to capture more of the demand curve.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Information Goods.

REAL-WORLD EXAMPLES

- Microsoft 365: Personal · Family · Business Basic · Standard · Premium · E3 · E5.
- Adobe Creative Cloud: Photography · Single App · All Apps · Teams · Enterprise.
- Spotify: Free · Individual · Duo · Family · Student.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Versioning Strategy contributes a clearer, more transferable framing of the dynamic it names — Multiple versions at different prices — let customers self-sort. Its analytical contribution is the explicit recognition that identify customer segments with different willingness to pay.

RELATED FRAMEWORKS IN THIS COURSE

Price Discrimination · 2.4.4 | Freemium Model · 2.3.2 | Subscription Economy · 2.3.3

SOURCES (HARVARD REFERENCING STYLE)

Shapiro, C. and Varian, H.R. (1999) Information Rules. Boston, MA: Harvard Business School Press, ch. 3.

Sundararajan, A. (2004) 'Nonlinear pricing of information goods', Management Science, 50(12), pp. 1660–1673.

Tellis, G.J. (1986) 'Beyond the many faces of price', Journal of Marketing, 50(4), pp. 146–160.

Price Discrimination

SLIBRARY 08 · SHEET 4 / 5

First-degree (personalised) · Second-degree (volume) · Third-degree (group).

PROFESSIONAL DEFINITION

The practice of charging different prices to different customers — or for different units of the same good — based on willingness to pay, without corresponding differences in cost. Pigou's classic three-degree taxonomy is the standard reference frame.

WHO DEVELOPED IT

Arthur Cecil Pigou, The Economics of Welfare (1920).

WHY IT WAS DEVELOPED

Extensively formalised in industrial-organisation theory; revitalised in the digital era because data and technology make finer-grained discrimination possible.

FIGURE 1 · PRICE DISCRIMINATION

Figure 1. Price Discrimination: First-degree (personalised) · Second-degree (volume) · Third-degree (group).

HOW IT IS USED

in pricing strategy when meaningful WTP heterogeneity is visible in the data.

WHEN IT IS USED

in regulatory and ethical analysis — first-degree discrimination raises fairness concerns.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Information Goods.

REAL-WORLD EXAMPLES

- Airline pricing: textbook second- and third-degree (refundability, advance purchase, frequent flyer).
- Amazon dynamic pricing: ~2.5m price changes per day across the catalogue (2024).
- Uber surge pricing: real-time, location-and-time-based — approaching first-degree in spirit.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Price Discrimination contributes a clearer, more transferable framing of the dynamic it names — First-degree (personalised) · Second-degree (volume) · Third-degree (group). Its analytical contribution is the explicit recognition that first-degree: each customer pays the maximum they would pay (perfect personalisation).

RELATED FRAMEWORKS IN THIS COURSE

Versioning Strategy · 2.4.3 | Freemium Model · 2.3.2 | GDPR & Data Rights (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

Pigou, A.C. (1920) The Economics of Welfare. London: Macmillan.

Tirole, J. (1988) The Theory of Industrial Organization. Cambridge, MA: MIT Press, ch. 3.

Shapiro, C. and Varian, H.R. (1999) Information Rules. Boston, MA: Harvard Business School Press.

Lock-In & Switching Costs

Contractual · Data · Skill · Network — the four sources of stickiness.

PROFESSIONAL DEFINITION

The mechanisms — contractual, technical, behavioural and economic — that make it costly for a customer to switch from one provider to another, thereby giving the incumbent provider durable pricing power.

WHO DEVELOPED IT

Foundational treatment in Shapiro and Varian, Information Rules (1999, ch.5).

WHY IT WAS DEVELOPED

Theoretical roots in Klemperer ('Markets with consumer switching costs', QJE, 1987). Re-examined for platform contexts by Farrell and Klemperer (Handbook of IO, 2007).

FIGURE 1 · LOCK-IN & SWITCHING COSTS

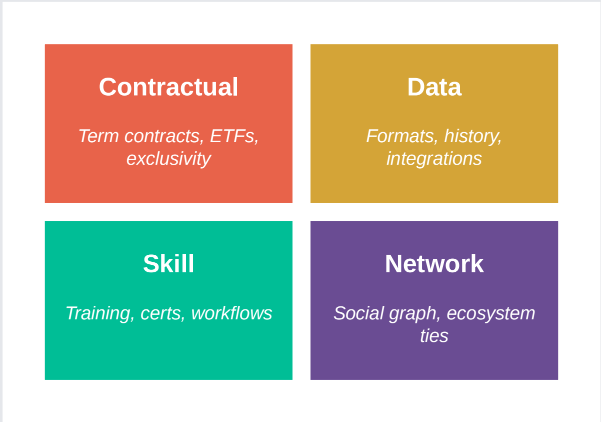

Figure 1. Lock-In & Switching Costs: Contractual · Data · Skill · Network — the four sources of stickiness.

HOW IT IS USED

when assessing the durability of a competitive position.

WHEN IT IS USED

when designing customer-acquisition and retention strategy.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Information Goods.

REAL-WORLD EXAMPLES

- AWS: data egress fees; bespoke services (Lambda, DynamoDB) re-engineered to leave — classic data + skill lock-in.
- Apple ecosystem: iMessage + iCloud + AirDrop + Continuity — multi-source lock-in.
- Bloomberg Terminal: keyboards, hotkeys, IB Chat — skill + network lock-in at $25k/seat.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Lock-In & Switching Costs contributes a clearer, more transferable framing of the dynamic it names — Contractual · Data · Skill · Network — the four sources of stickiness. Its analytical contribution is the explicit recognition that contractual lock-in: term contracts, early-termination fees, exclusivity clauses.

RELATED FRAMEWORKS IN THIS COURSE

Tipping Point · Winner-Takes-All · 2.2.5 | Subscription Economy · 2.3.3 | Platform Liability (Unit 5)

SOURCES (HARVARD REFERENCING STYLE)

Shapiro, C. and Varian, H.R. (1999) Information Rules. Boston, MA: Harvard Business School Press, ch. 5.

Klemperer, P. (1987) 'Markets with consumer switching costs', Quarterly Journal of Economics, 102(2), pp. 375–394.

Farrell, J. and Klemperer, P. (2007) 'Coordination and lock-in', in Handbook of Industrial Organization, vol. 3. Amsterdam: North-Holland.